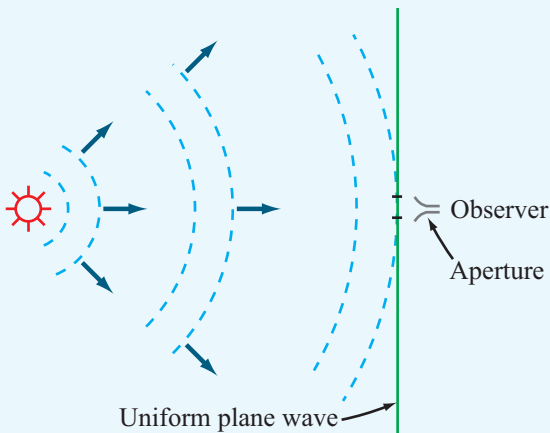
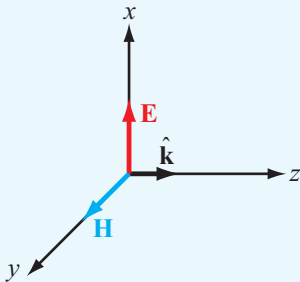


(a) Spherical wave

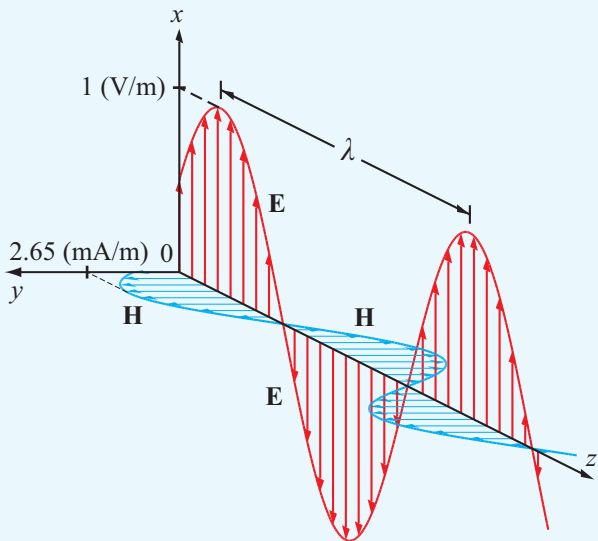


(b) Plane-wave approximation

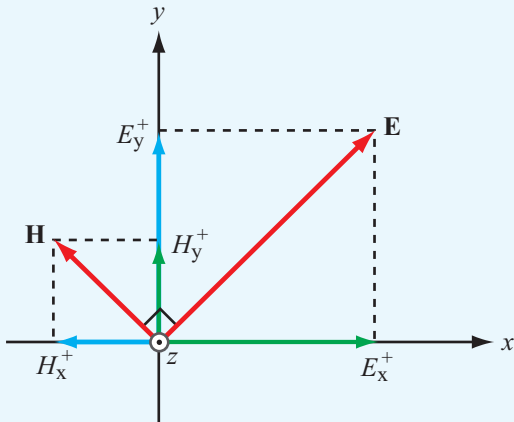
**Figure 2-1:** Waves radiated by an EM source, such as a lightbulb or an antenna, have spherical wavefronts, as in (a); to a distant observer, however, the wavefront across the observer's aperture appears approximately planar, as in (b).



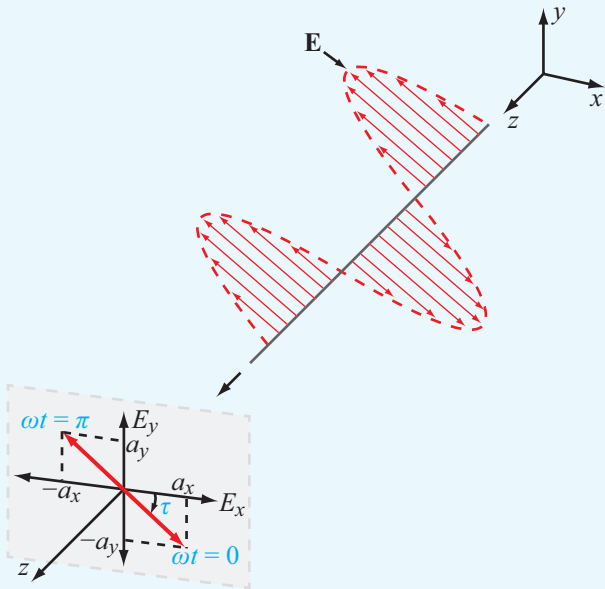
**Figure 2-2:** A transverse electromagnetic (TEM) wave propagating in the direction  $\hat{\mathbf{k}} = \hat{\mathbf{z}}$ . For all TEM waves,  $\hat{\mathbf{k}}$  is parallel to  $\mathbf{E} \times \mathbf{H}$ .



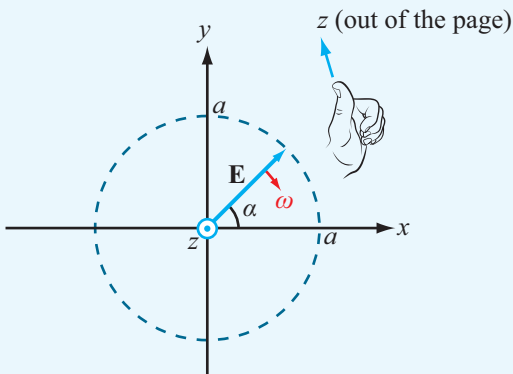
**Figure 2-3:** Spatial variations of  $\mathbf{E}$  and  $\mathbf{H}$  at  $t = 0$  for the plane wave defined by Eq. (2.33).



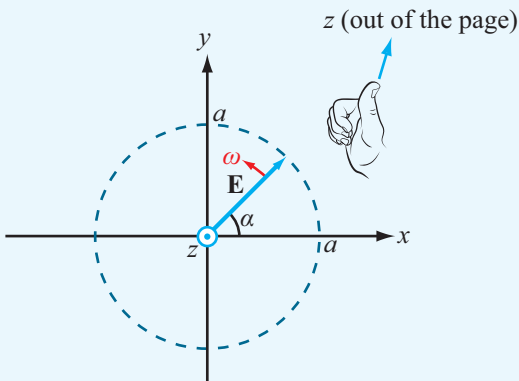
**Figure 2-4:** The wave  $(\mathbf{E}, \mathbf{H})$  is equivalent to the sum of two waves, one with fields  $(E_x^+, H_y^+)$  and another with  $(E_y^+, H_x^+)$ , with both traveling in the  $+z$  direction.



**Figure 2-5:** Linearly polarized wave traveling in the  $+z$  direction.

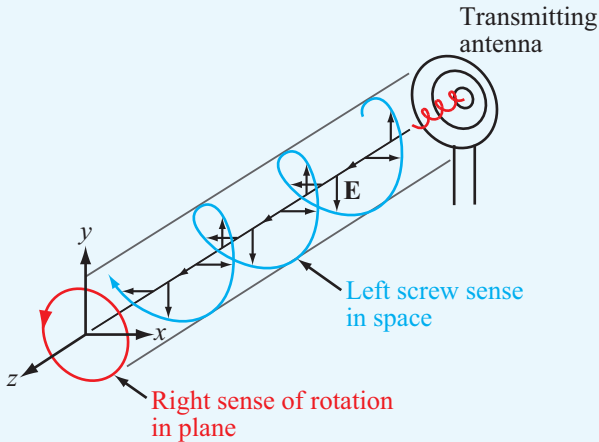


(a) LHC polarization

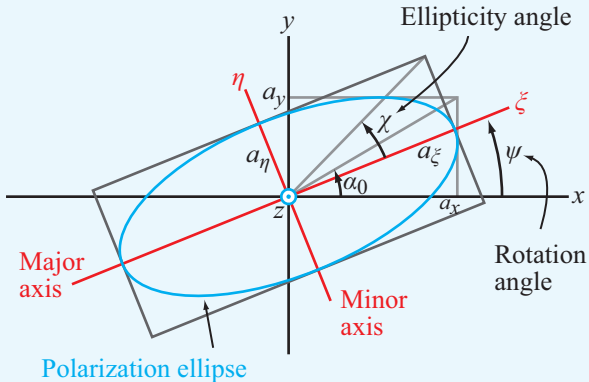


(b) RHC polarization

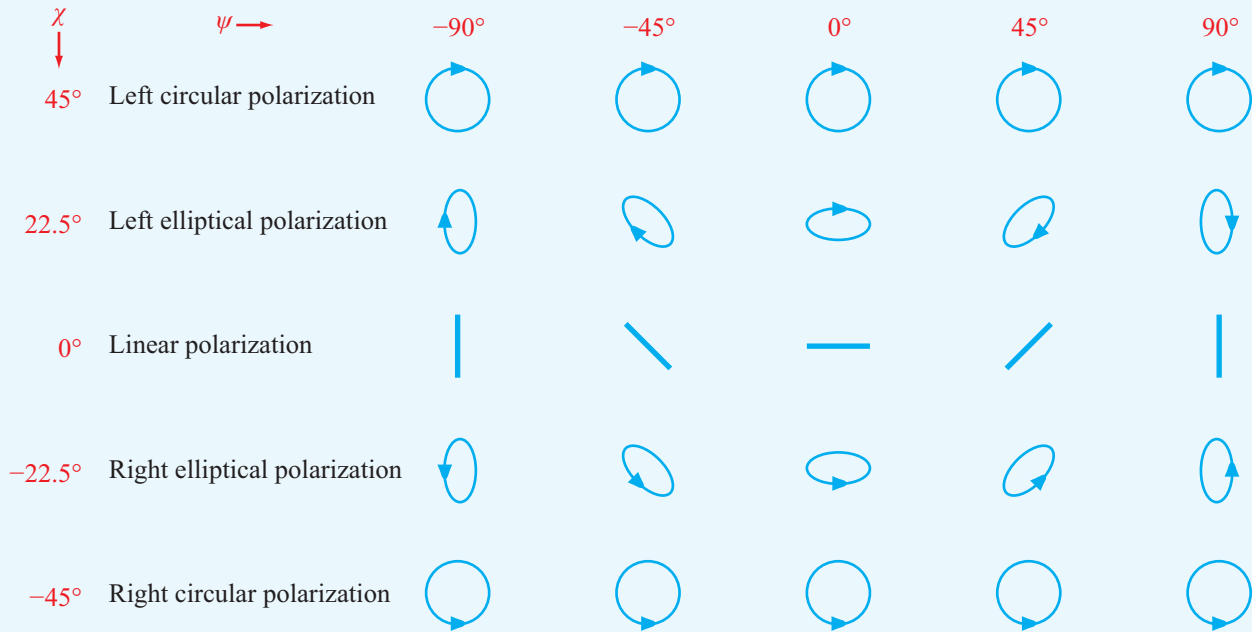
**Figure 2-6:** Circularly polarized plane waves propagating in the  $+z$  direction (out of the page).



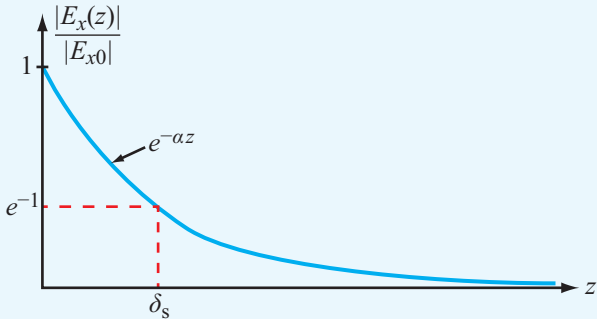
**Figure 2-7:** Right-hand circularly polarized wave radiated by a helical antenna.



**Figure 2-8:** Polarization ellipse in the  $x$ - $y$  plane, with the wave traveling in the  $z$  direction (out of the page).



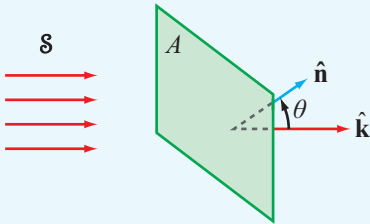
**Figure 2-9:** Polarization states for various combinations of the polarization angles  $(\psi, \chi)$  for a wave traveling out of the page.<sup>†</sup>



**Figure 2-10:** Attenuation of the magnitude of  $E_x(z)$  with distance  $z$ . The skin depth  $\delta_s$  is the value of  $z$  at which  $|E_x(z)|/|E_{x0}| = e^{-1}$ , or  $z = \delta_s = 1/\alpha$ .

**Table 2-1:** Expressions for  $\alpha$ ,  $\beta$ ,  $\eta_c$ ,  $u_p$ , and  $\lambda$  for various types of nonmagnetic media.<sup>†</sup>

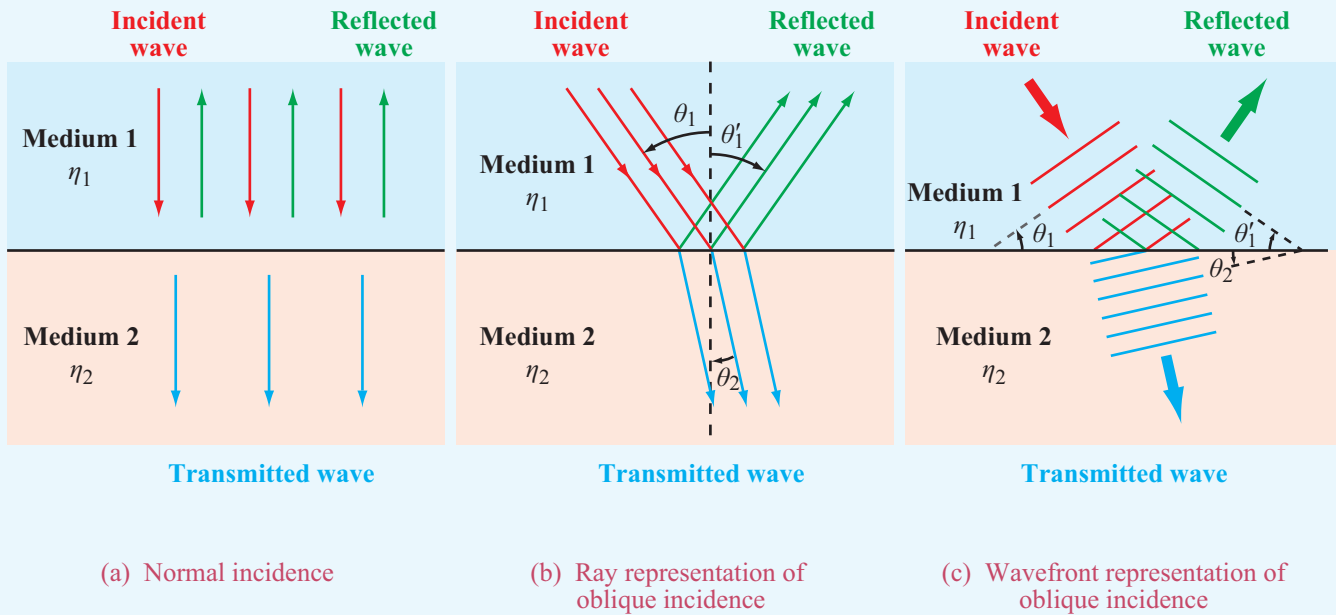
|                                                                                                                                                                                                                 | Any Medium                                                                                                                                     | Lossless Medium<br>( $\sigma = 0$ )       | Low-Loss Medium<br>( $\epsilon''/\epsilon' \ll 1$ ) | Good Conductor<br>( $\epsilon''/\epsilon' \gg 1$ ) | Units        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------|----------------------------------------------------|--------------|
| $\alpha =$                                                                                                                                                                                                      | $\omega \left[ \frac{\mu_0 \epsilon' \epsilon_0}{2} \left[ \sqrt{1 + \left( \frac{\epsilon''}{\epsilon'} \right)^2} - 1 \right] \right]^{1/2}$ | 0                                         | $\frac{\pi \epsilon''}{\lambda_0 \sqrt{\epsilon'}}$ | $\frac{\pi \sqrt{2 \epsilon''}}{\lambda_0}$        | (Np/m)       |
| $\beta =$                                                                                                                                                                                                       | $\omega \left[ \frac{\mu_0 \epsilon' \epsilon_0}{2} \left[ \sqrt{1 + \left( \frac{\epsilon''}{\epsilon'} \right)^2} + 1 \right] \right]^{1/2}$ | $\frac{2\pi \sqrt{\epsilon'}}{\lambda_0}$ | $\frac{2\pi \sqrt{\epsilon'}}{\lambda_0}$           | $\frac{\pi \sqrt{2 \epsilon''}}{\lambda_0}$        | (rad/m)      |
| $\eta_c =$                                                                                                                                                                                                      | $\sqrt{\frac{\mu_0}{\epsilon' \epsilon_0}} \left( 1 - j \frac{\epsilon''}{\epsilon'} \right)^{-1/2}$                                           | $\frac{\eta_0}{\sqrt{\epsilon'}}$         | $\frac{\eta_0}{\sqrt{\epsilon'}}$                   | $\frac{(1 + j) \eta_0}{\sqrt{2 \epsilon''}}$       | ( $\Omega$ ) |
| $u_p =$                                                                                                                                                                                                         | $\omega / \beta$                                                                                                                               | $c / \sqrt{\epsilon'}$                    | $c / \sqrt{\epsilon'}$                              | $c \sqrt{2 / \epsilon''}$                          | (m/s)        |
| $\lambda =$                                                                                                                                                                                                     | $2\pi / \beta = u_p / f$                                                                                                                       | $u_p / f$                                 | $u_p / f$                                           | $u_p / f$                                          | (m)          |
| Notes: In practice, a material is considered a low-loss medium if $\epsilon''/\epsilon' < 0.01$ and a good conducting medium if $\epsilon''/\epsilon' > 100$ ; $c = 3 \times 10^8$ m/s; $\eta_0 = 377 \Omega$ . |                                                                                                                                                |                                           |                                                     |                                                    |              |



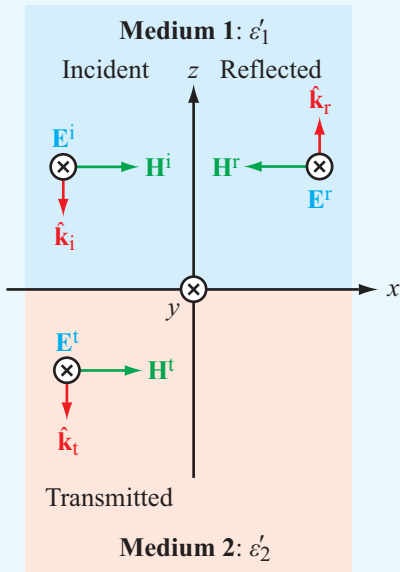
**Figure 2-11:** EM power flow through an aperture.

**Table 2-2: Power ratios in natural numbers and in decibels.**

| $G$       | $G$ [dB] |
|-----------|----------|
| $10^x$    | $10x$ dB |
| 4         | 6 dB     |
| 2         | 3 dB     |
| 1         | 0 dB     |
| 0.5       | -3 dB    |
| 0.25      | -6 dB    |
| 0.1       | -10 dB   |
| $10^{-3}$ | -30 dB   |



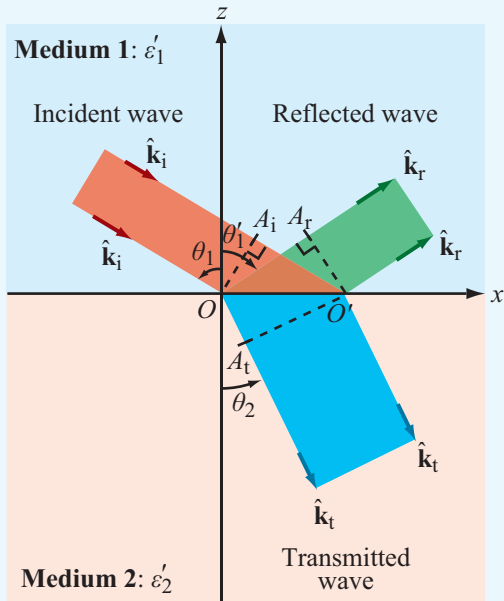
**Figure 2-12:** Ray representation of wave reflection and transmission at (a) normal incidence and (b) oblique incidence, and (c) wavefront representation of oblique incidence.



**Figure 2-13:** Two dielectric media separated by the  $x$ - $y$  plane.

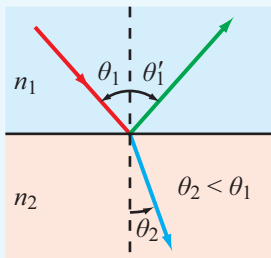
**Table 2-3:** Expressions for EM fields in lossless and lossy media under normal incidence.<sup>†</sup>

| Lossless Media                                                                                                                                                                | Lossy Media                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathbf{E}^i = \hat{\mathbf{y}} E_0^i e^{jk_1 z}, \quad \mathbf{H}^i = \hat{\mathbf{x}} \frac{E_0^i}{\eta_1} e^{jk_1 z}$                                                     | $\mathbf{E}^i = \hat{\mathbf{y}} E_0^i e^{\gamma_1 z}, \quad \mathbf{H}^i = \hat{\mathbf{x}} \frac{E_0^i}{\eta_{c1}} e^{\gamma_1 z}$                                          |
| $\mathbf{E}^r = \hat{\mathbf{y}} \rho E_0^i e^{-jk_1 z}, \quad \mathbf{H}^r = -\hat{\mathbf{x}} \rho \frac{E_0^i}{\eta_1} e^{-jk_1 z}$                                        | $\mathbf{E}^r = \hat{\mathbf{y}} \rho E_0^i e^{-\gamma_1 z}, \quad \mathbf{H}^r = -\hat{\mathbf{x}} \rho \frac{E_0^i}{\eta_{c1}} e^{-\gamma_1 z}$                             |
| $\mathbf{E}^t = \hat{\mathbf{y}} \tau E_0^i e^{jk_2 z}, \quad \mathbf{H}^t = \hat{\mathbf{x}} \tau \frac{E_0^i}{\eta_2} e^{jk_2 z}$                                           | $\mathbf{E}^t = \hat{\mathbf{y}} \tau E_0^i e^{\gamma_2 z}, \quad \mathbf{H}^t = \hat{\mathbf{x}} \tau \frac{E_0^i}{\eta_{c2}} e^{\gamma_2 z}$                                |
| <p>Phase matching: <math>(\mathbf{E}^i + \mathbf{E}^r) _{z=0} = \mathbf{E}^t _{z=0}</math></p> <p><math>(\mathbf{H}^i + \mathbf{H}^r) _{z=0} = \mathbf{H}^t _{z=0}</math></p> | <p>Phase matching: <math>(\mathbf{E}^i + \mathbf{E}^r) _{z=0} = \mathbf{E}^t _{z=0}</math></p> <p><math>(\mathbf{H}^i + \mathbf{H}^r) _{z=0} = \mathbf{H}^t _{z=0}</math></p> |
| $\rho = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}, \quad \tau = 1 + \rho$                                                                                                       | $\rho = \frac{\eta_{c2} - \eta_{c1}}{\eta_{c2} + \eta_{c1}}, \quad \tau = 1 + \rho$                                                                                           |
| $k_1 = \frac{2\pi}{\lambda_0} \sqrt{\epsilon'_1}, \quad k_2 = \frac{2\pi}{\lambda_0} \sqrt{\epsilon'_2}$                                                                      | $\gamma_1 = \alpha_1 + j\beta_1, \quad \gamma_2 = \alpha_2 + j\beta_2$                                                                                                        |
| $\eta_1 = \frac{\eta_0}{\sqrt{\epsilon'_1}}, \quad \eta_2 = \frac{\eta_0}{\sqrt{\epsilon'_2}}$                                                                                | $\eta_{c1} = \frac{\eta_0}{\sqrt{\epsilon_1}}, \quad \eta_{c2} = \frac{\eta_0}{\sqrt{\epsilon_2}}$                                                                            |
| <p>Notes: <math>\epsilon_1 = \epsilon'_1 - j\epsilon''_1</math>; <math>\epsilon_2 = \epsilon'_2 - j\epsilon''_2</math>.</p>                                                   |                                                                                                                                                                               |



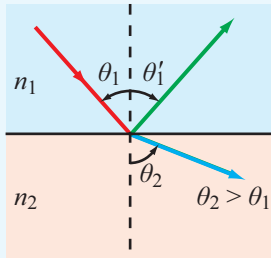
**Figure 2-14:** Wave reflection and refraction at a planar boundary between different media.

### Inward refraction



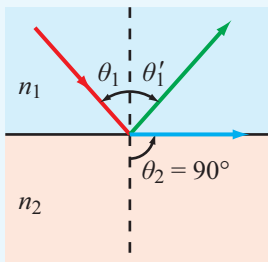
(a)  $n_1 < n_2$

### Outward refraction



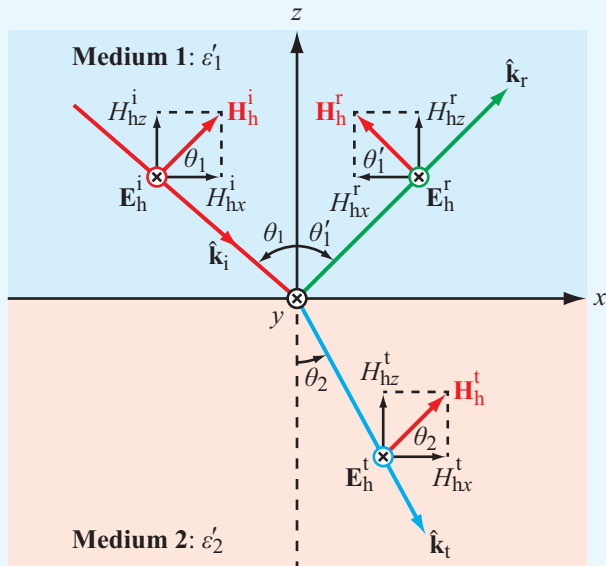
(b)  $n_1 > n_2$

### No transmission

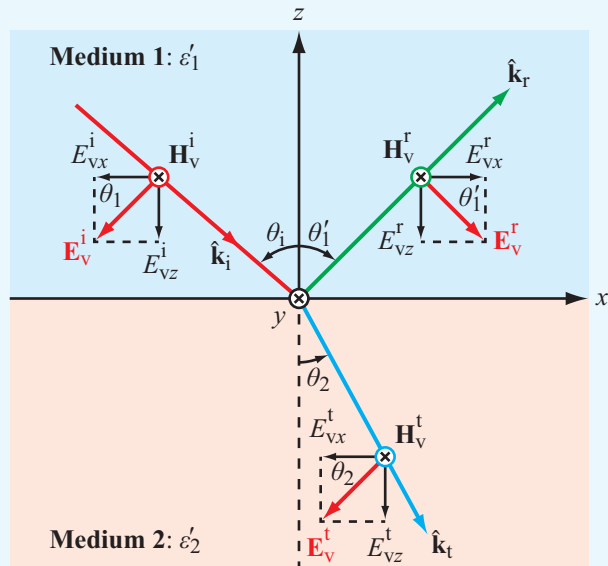


(c)  $n_1 > n_2$  and  $\theta_1 = \theta_c$

**Figure 2-15:** Snell's laws state that  $\theta'_1 = \theta_1$  and  $\sin \theta_2 = (n_1/n_2) \sin \theta_1$ . Refraction is (a) inward if  $n_1 < n_2$  and (b) outward if  $n_1 > n_2$ ; and (c) the refraction angle is  $90^\circ$  if  $n_1 > n_2$  and  $\theta_1$  is equal to or greater than the critical angle  $\theta_c = \sin^{-1}(n_2/n_1)$ .



(a) Horizontal polarization

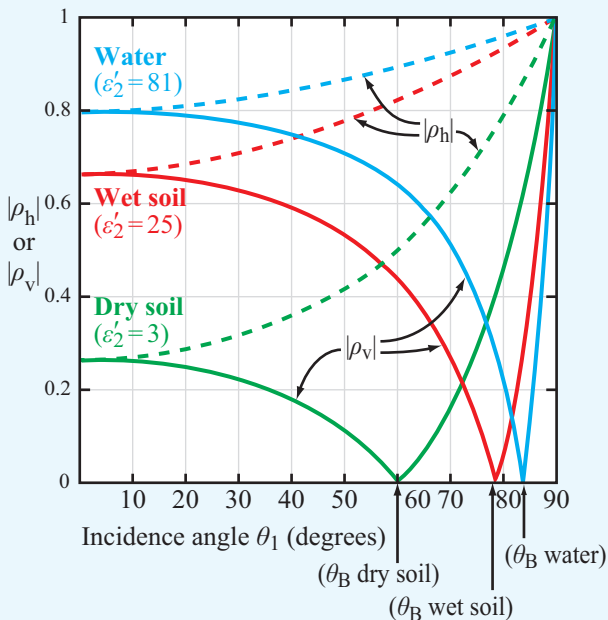


(b) Vertical polarization

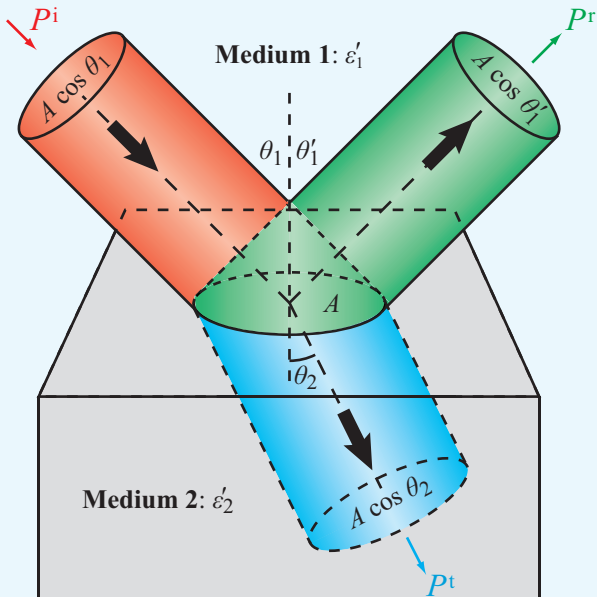
**Figure 2-16:** The plane of incidence is the plane containing the direction of wave travel,  $\hat{k}_i$ , and the surface normal to the boundary. In the present case the plane of incidence containing  $\hat{k}_i$  and  $\hat{z}$  coincides with the plane of the paper. A wave is (a) perpendicularly polarized (also called **horizontally polarized**) when its electric field vector is perpendicular to the plane of incidence and (b) parallel polarized (also called **vertically polarized**) when its electric field vector lies in the plane of incidence.

**Table 2-4: Expressions for EM fields in lossless and lossy\* media under normal incidence.<sup>†</sup>**

| Horizontal Polarization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Vertical Polarization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathbf{E}_h^i = \hat{\mathbf{y}} E_{h0}^i e^{-jk_1(x \sin \theta_1 - z \cos \theta_1)}$ $\mathbf{H}_h^i = (\hat{\mathbf{x}} \cos \theta_1 + \hat{\mathbf{z}} \sin \theta_1) \frac{E_{h0}^i}{\eta_1} e^{-jk_1(x \sin \theta_1 - z \cos \theta_1)}$ $\mathbf{E}_h^r = \hat{\mathbf{y}} \rho_h E_{h0}^i e^{-jk_1(x \sin \theta_1 + z \cos \theta_1)}$ $\mathbf{H}_h^r = (-\hat{\mathbf{x}} \cos \theta_1 + \hat{\mathbf{z}} \sin \theta_1) \rho_h \frac{E_{h0}^i}{\eta_1} e^{-jk_1(x \sin \theta_1 + z \cos \theta_1)}$ $\mathbf{E}_h^t = \hat{\mathbf{y}} \tau_h E_{h0}^i e^{-jk_2(x \sin \theta_2 - z \cos \theta_2)}$ $\mathbf{H}_h^t = (\hat{\mathbf{x}} \cos \theta_2 + \hat{\mathbf{z}} \sin \theta_2) \tau_h \frac{E_{h0}^i}{\eta_2} e^{-jk_2(x \sin \theta_2 - z \cos \theta_2)}$ $\rho_h = \frac{\eta_2 \cos \theta_1 - \eta_1 \cos \theta_2}{\eta_2 \cos \theta_1 + \eta_1 \cos \theta_2} = \frac{\cos \theta_1 - \sqrt{(\epsilon'_2/\epsilon'_1) - \sin^2 \theta_1}}{\cos \theta_1 + \sqrt{(\epsilon'_2/\epsilon'_1) - \sin^2 \theta_1}}$ $\tau_h = 1 + \rho_h$ | $\mathbf{E}_v^i = (-\hat{\mathbf{x}} \cos \theta_1 - \hat{\mathbf{z}} \sin \theta_1) E_{v0}^i e^{-jk_1(x \sin \theta_1 - z \cos \theta_1)}$ $\mathbf{H}_v^i = \hat{\mathbf{y}} \frac{E_{v0}^i}{\eta_1} e^{-jk_1(x \sin \theta_1 - z \cos \theta_1)}$ $\mathbf{E}_v^r = (\hat{\mathbf{x}} \cos \theta_1 - \hat{\mathbf{z}} \sin \theta_1) \rho_v E_{v0}^i e^{-jk_1(x \sin \theta_1 + z \cos \theta_1)}$ $\mathbf{H}_v^r = \hat{\mathbf{y}} \rho_v \frac{E_{v0}^i}{\eta_1} e^{-jk_1(x \sin \theta_1 + z \cos \theta_1)}$ $\mathbf{E}_v^t = (-\hat{\mathbf{x}} \cos \theta_2 - \hat{\mathbf{z}} \sin \theta_2) \tau_v E_{v0}^i e^{-jk_2(x \sin \theta_2 - z \cos \theta_2)}$ $\mathbf{H}_v^t = \hat{\mathbf{y}} \tau_v \frac{E_{v0}^i}{\eta_1} e^{-jk_2(x \sin \theta_2 - z \cos \theta_2)}$ $\rho_v = \frac{\eta_2 \cos \theta_2 - \eta_1 \cos \theta_1}{\eta_2 \cos \theta_2 + \eta_1 \cos \theta_1} = \frac{\left[ \frac{\epsilon'_2}{\epsilon'_1} - \sin^2 \theta_1 \right]^{1/2} - \left( \frac{\epsilon'_2}{\epsilon'_1} \right) \cos \theta_1}{\left[ \frac{\epsilon'_2}{\epsilon'_1} - \sin^2 \theta_1 \right]^{1/2} + \left( \frac{\epsilon'_2}{\epsilon'_1} \right) \cos \theta_1}$ $\tau_v = (1 + \rho_v) \frac{\cos \theta_1}{\cos \theta_2}$ |
| $k_1 = \frac{2\pi}{\lambda_0} \sqrt{\epsilon_1}$ $k_2 = \frac{2\pi}{\lambda} \sqrt{\epsilon_2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\eta_1 = \frac{\eta_0}{\sqrt{\epsilon_1}}$ $\eta_2 = \frac{\eta_0}{\sqrt{\epsilon_2}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $k_2 \sin \theta_2 = k_1 \sin \theta_1$ $\cos \theta_2 = \left[ 1 - \left( \frac{k_1}{k_2} \sin \theta_1 \right)^2 \right]^{1/2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p>*Notes: (1) Lossless medium: <math>\epsilon = \epsilon'</math>; (2) lossy medium: <math>\epsilon = \epsilon' - j\epsilon''</math> and <math>jk</math> should be replaced with <math>\gamma = j(2\pi\sqrt{\epsilon' - j\epsilon''})/\lambda_0</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |



**Figure 2-17:** Plots for  $|\rho_h|$  and  $|\rho_v|$  as a function of  $\theta_1$  for a dry soil surface, a wet-soil surface, and a water surface. For each surface,  $|\rho_v| = 0$  at the Brewster angle.



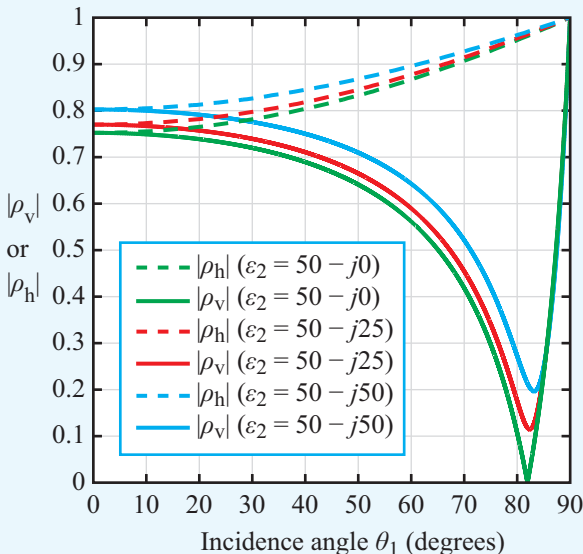
**Figure 2-18:** Reflection and transmission of an incident circular beam illuminating a spot of size  $A$  on the interface.

**Table 2-5:** Expressions for  $\rho$ ,  $\tau$ ,  $\Gamma$ , and  $\mathbb{T}$  for wave incidence from a lossless medium with intrinsic impedance  $\eta_1$  onto a second lossless medium with intrinsic impedance  $\eta_2$ . Angles  $\theta_1$  and  $\theta_2$  are the angles of incidence and transmission, respectively.<sup>†</sup>

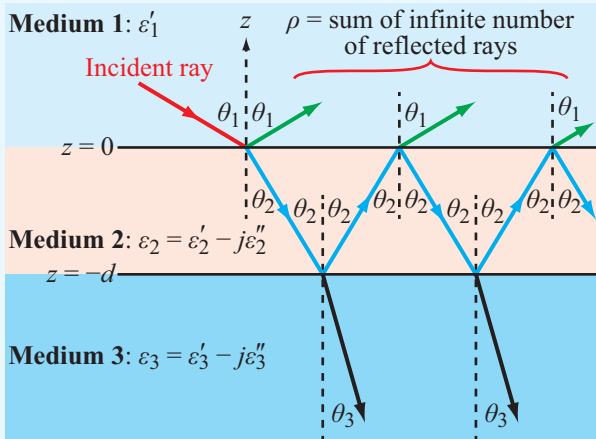
| Property                                                                                                                                              | Normal Incidence<br>$\theta_1 = \theta_2 = 0$                | Horizontal Polarization                                                                                    | Vertical Polarization                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Reflection coefficient                                                                                                                                | $\rho = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$             | $\rho_h = \frac{\eta_2 \cos \theta_1 - \eta_1 \cos \theta_2}{\eta_2 \cos \theta_1 + \eta_1 \cos \theta_2}$ | $\rho_v = \frac{\eta_2 \cos \theta_2 - \eta_1 \cos \theta_1}{\eta_2 \cos \theta_2 + \eta_1 \cos \theta_1}$ |
| Transmission coefficient                                                                                                                              | $\tau = \frac{2\eta_2}{\eta_2 + \eta_1}$                     | $\tau_h = \frac{2\eta_2 \cos \theta_1}{\eta_2 \cos \theta_1 + \eta_1 \cos \theta_2}$                       | $\tau_v = \frac{2\eta_2 \cos \theta_1}{\eta_2 \cos \theta_2 + \eta_1 \cos \theta_1}$                       |
| Relation of $\rho$ to $\tau$                                                                                                                          | $\tau = 1 + \rho$                                            | $\tau_h = 1 + \rho_h$                                                                                      | $\tau_v = (1 + \rho_v) \frac{\cos \theta_1}{\cos \theta_2}$                                                |
| Reflectivity                                                                                                                                          | $\Gamma =  \rho ^2$                                          | $\Gamma^h =  \rho_h ^2$                                                                                    | $\Gamma^v =  \rho_v ^2$                                                                                    |
| Transmissivity                                                                                                                                        | $\mathbb{T} =  \tau ^2 \left( \frac{\eta_1}{\eta_2} \right)$ | $\mathbb{T}^h =  \tau_h ^2 \frac{\eta_1 \cos \theta_2}{\eta_2 \cos \theta_1}$                              | $\mathbb{T}^v =  \tau_v ^2 \frac{\eta_1 \cos \theta_2}{\eta_2 \cos \theta_1}$                              |
| Relation of $\Gamma$ to $\mathbb{T}$                                                                                                                  | $\mathbb{T} = 1 - \Gamma$                                    | $\mathbb{T}^h = 1 - \Gamma^h$                                                                              | $\mathbb{T}^v = 1 - \Gamma^v$                                                                              |
| Notes: $\sin \theta_2 = \sqrt{\epsilon'_1/\epsilon'_2} \sin \theta_1$ ; $\eta_1 = \eta_0/\sqrt{\epsilon'_1}$ ; $\eta_2 = \eta_0/\sqrt{\epsilon'_2}$ . |                                                              |                                                                                                            |                                                                                                            |

**Table 2-6: Oblique incidence in a lossless medium onto a lossy medium.<sup>†</sup>**

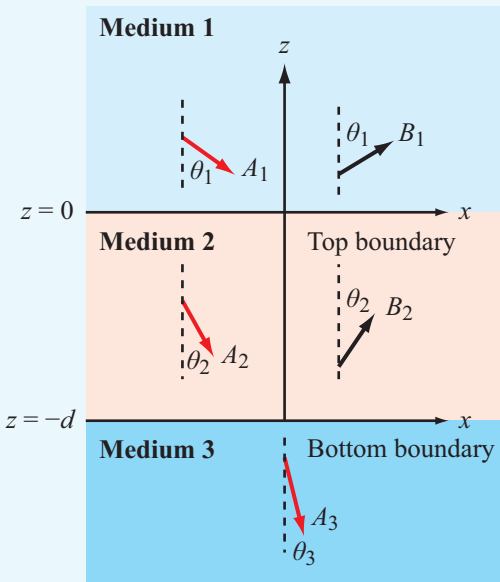
| Horizontal Polarization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Vertical Polarization                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathbf{E}_h^i = \hat{\mathbf{y}} E_{h0}^i e^{-jk_1(x \sin \theta_1 - z \cos \theta_1)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\mathbf{E}_v^i = (-\hat{\mathbf{x}} \cos \theta_1 - \hat{\mathbf{z}} \sin \theta_1) E_{v0}^i e^{-jk_1(x \sin \theta_1 - z \cos \theta_1)}$            |
| $\mathbf{H}_h^i = (\hat{\mathbf{x}} \cos \theta_1 + \hat{\mathbf{z}} \sin \theta_1) \frac{E_{h0}^i}{\eta_1} e^{-jk_1(x \sin \theta_1 - z \cos \theta_1)}$                                                                                                                                                                                                                                                                                                                                                                                                      | $\mathbf{H}_v^i = \hat{\mathbf{y}} \frac{E_{v0}^i}{\eta_1} e^{-jk_1(x \sin \theta_1 - z \cos \theta_1)}$                                               |
| $\mathbf{E}_h^r = \hat{\mathbf{y}} \rho_h E_{h0}^i e^{-jk_1(x \sin \theta_1 + z \cos \theta_1)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\mathbf{E}_v^r = (\hat{\mathbf{x}} \cos \theta_1 - \hat{\mathbf{z}} \sin \theta_1) \rho_v E_{v0}^i e^{-jk_1(x \sin \theta_1 + z \cos \theta_1)}$      |
| $\mathbf{H}_h^r = (-\hat{\mathbf{x}} \cos \theta_1 + \hat{\mathbf{z}} \sin \theta_1) \rho_h \frac{E_{h0}^i}{\eta_1} e^{-jk_1(x \sin \theta_1 + z \cos \theta_1)}$                                                                                                                                                                                                                                                                                                                                                                                              | $\mathbf{H}_v^r = \hat{\mathbf{y}} \rho_v \frac{E_{v0}^i}{\eta_1} e^{-jk_1(x \sin \theta_1 + z \cos \theta_1)}$                                        |
| $\mathbf{E}_h^t = \hat{\mathbf{y}} \tau_h E_{h0}^i e^{-\gamma_2(x \sin \theta_2 - z \cos \theta_2)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\mathbf{E}_v^t = (-\hat{\mathbf{x}} \cos \theta_2 - \hat{\mathbf{z}} \sin \theta_2) \tau_v E_{v0}^i e^{-\gamma_2(x \sin \theta_2 - z \cos \theta_2)}$ |
| $\mathbf{H}_h^t = (\hat{\mathbf{x}} \cos \theta_2 + \hat{\mathbf{z}} \sin \theta_2) \tau_h \frac{E_{h0}^i}{\eta_2} e^{-\gamma_2(x \sin \theta_2 - z \cos \theta_2)}$                                                                                                                                                                                                                                                                                                                                                                                           | $\mathbf{H}_v^t = \hat{\mathbf{y}} \tau_v \frac{E_{v0}^i}{\eta_1} e^{-\gamma_2(x \sin \theta_2 - z \cos \theta_2)}$                                    |
| $\rho_h = \frac{\eta_2 \cos \theta_1 - \eta_1 \cos \theta_2}{\eta_2 \cos \theta_1 + \eta_1 \cos \theta_2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\rho_v = \frac{\eta_2 \cos \theta_2 - \eta_1 \cos \theta_1}{\eta_2 \cos \theta_2 + \eta_1 \cos \theta_1}$                                             |
| $\tau_h = 1 + \rho_h$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\tau_v = (1 + \rho_v) \frac{\cos \theta_1}{\cos \theta_2}$                                                                                            |
| <p>Notes:</p> $k_1 = \frac{2\pi}{\lambda_0} \sqrt{\epsilon'_1} \qquad \eta_1 = \frac{\eta_0}{\sqrt{\epsilon'_1}} \qquad \eta_2 = \frac{\eta_0}{\sqrt{\epsilon'_2}}$ $\gamma_2 = \alpha_2 + j\beta_2 = j \frac{2\pi}{\lambda_0} \sqrt{\epsilon'_2} \qquad \gamma_2 \sin \theta_2 = k_1 \sin \theta_1 \qquad \epsilon_2 = \epsilon'_2 - j\epsilon''_2$ $\cos \theta_2 = \left[ 1 - \left( \frac{jk_1}{\gamma_2} \sin \theta_1 \right)^2 \right]^{1/2} = \left[ 1 - \left( \frac{\epsilon'_1}{\epsilon'_2 - j\epsilon''_2} \right) \sin^2 \theta_1 \right]^{1/2}$ |                                                                                                                                                        |



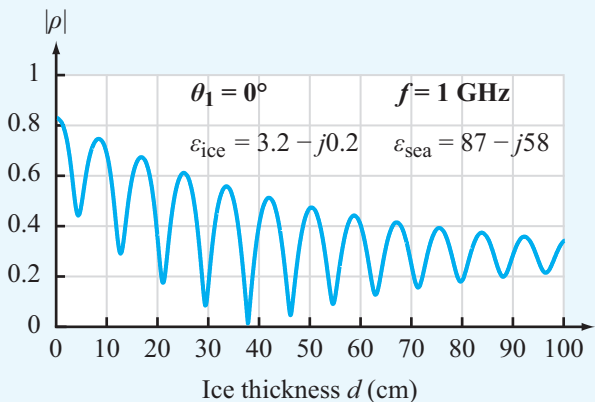
**Figure 2-19:** Angular plots of  $|\rho_v|$  and  $|\rho_h|$  for incidence in air upon a material with  $\epsilon_2 = \epsilon'_2 - j\epsilon''_2$ .



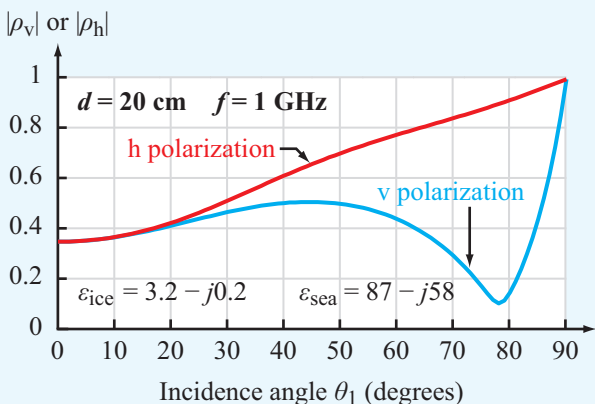
**Figure 2-20:** Multiple reflections in a two-layer composite.



**Figure 2-21:** Electric field amplitudes of downward and upward propagating waves in media 1, 2 and 3.

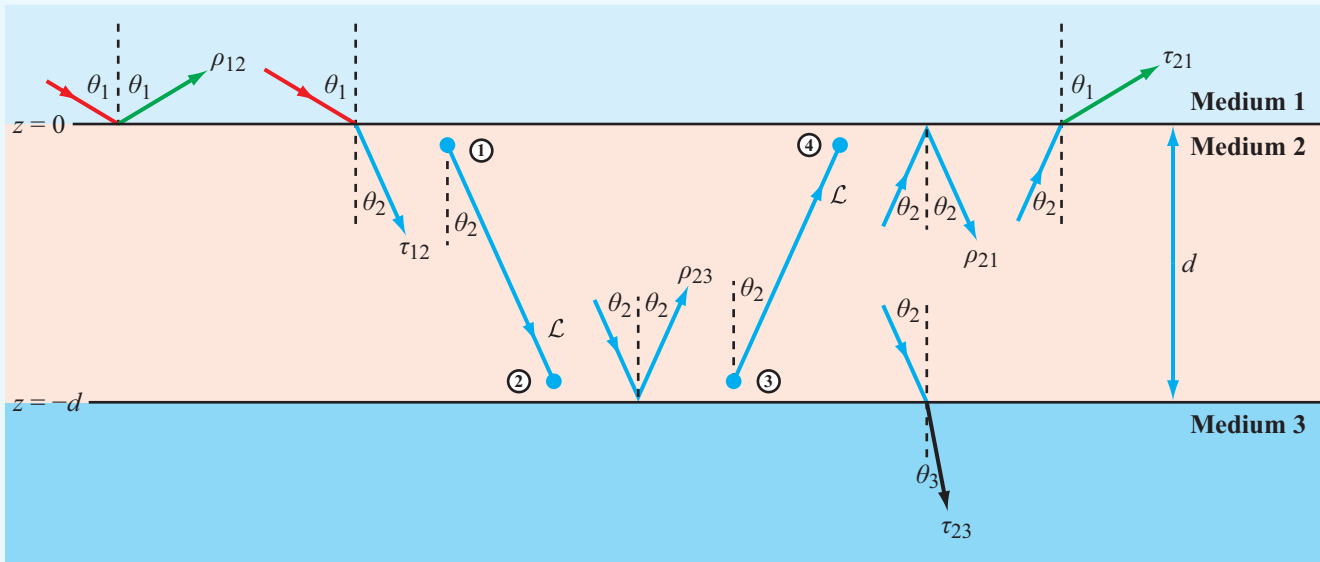


(a)  $|\rho|$  versus  $d$

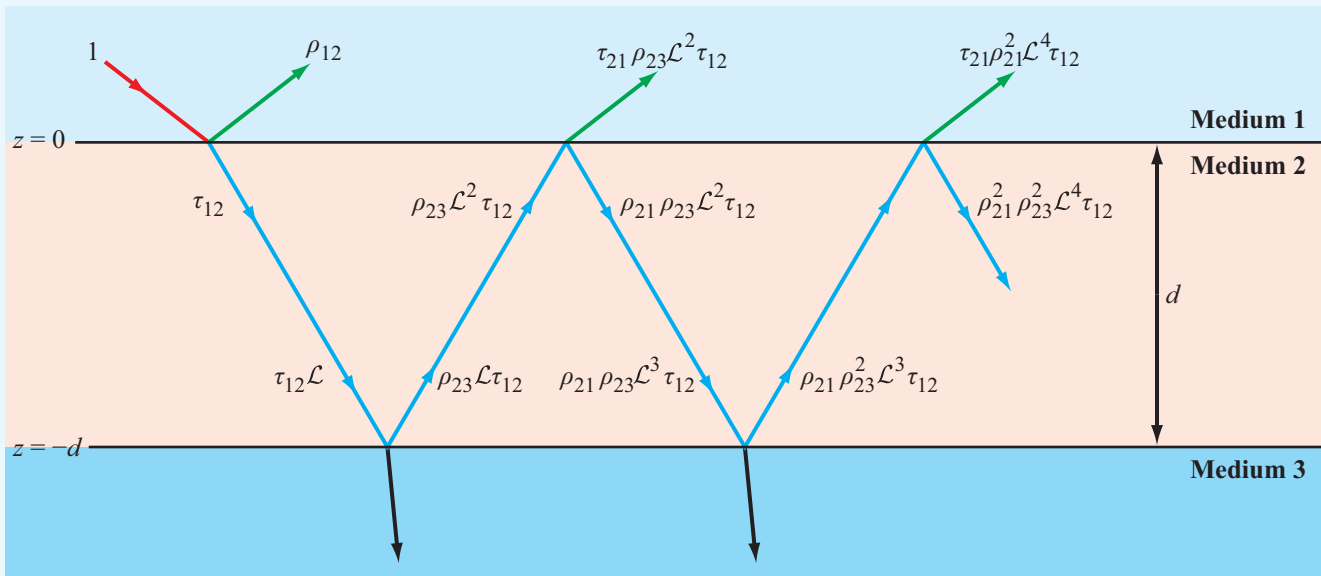


(b)  $|\rho_v|$  and  $|\rho_h|$  versus  $\theta$

**Figure 2-22:** Plots of (a)  $|\rho|$  as a function of ice-layer thickness  $d$  for normal incidence and (b)  $|\rho_v|$  and  $|\rho_h|$  as a function of incidence angle  $\theta_1$  for a 20 cm ice layer over the ocean surface.



**Figure 2-23:** Reflection, transmission, and propagation mechanisms.



**Figure 2-24:** Multiple reflection process.

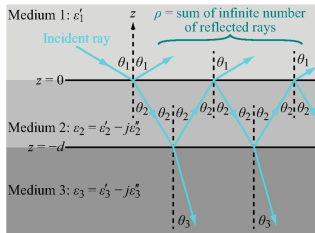
## Code 2.4

This module computes the reflection properties of a two-layer composite with planar boundaries. Medium 1 is air with  $\epsilon_1 = 1$ . The incidence angle in medium 1, and the frequency in GHz also are inputs. The reflection coefficient and reflectivity are plotted against the thickness of the top layer, for both h and v polarizations.

$$\epsilon_2 = \epsilon_2' - j\epsilon_2''$$

$$\epsilon_3 = \epsilon_3' - j\epsilon_3''$$

matlab code: [Refl\\_TwoLayerComposite.m](#)



Real Part:  $\epsilon_2'$ :

Imag Part:  $\epsilon_2''$ :

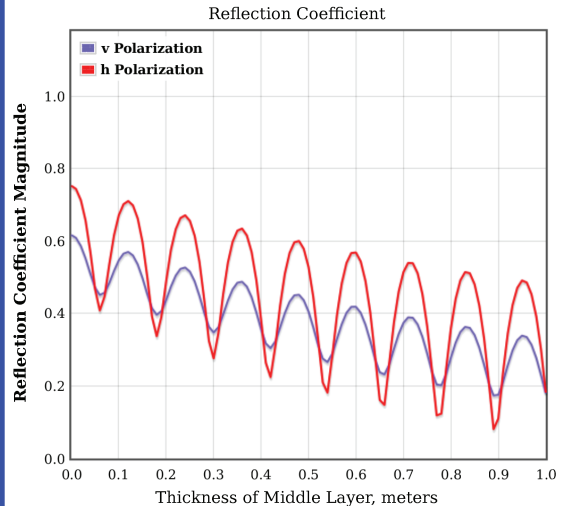
Real Part:  $\epsilon_3'$ :

Imag Part:  $\epsilon_3''$ :

Incidence Angle (deg):

Frequency (GHz):

## Reflection by Two-Layer Composite



☒ Plot Reflection Coefficient Magnitude  
☐ Plot Reflectivity